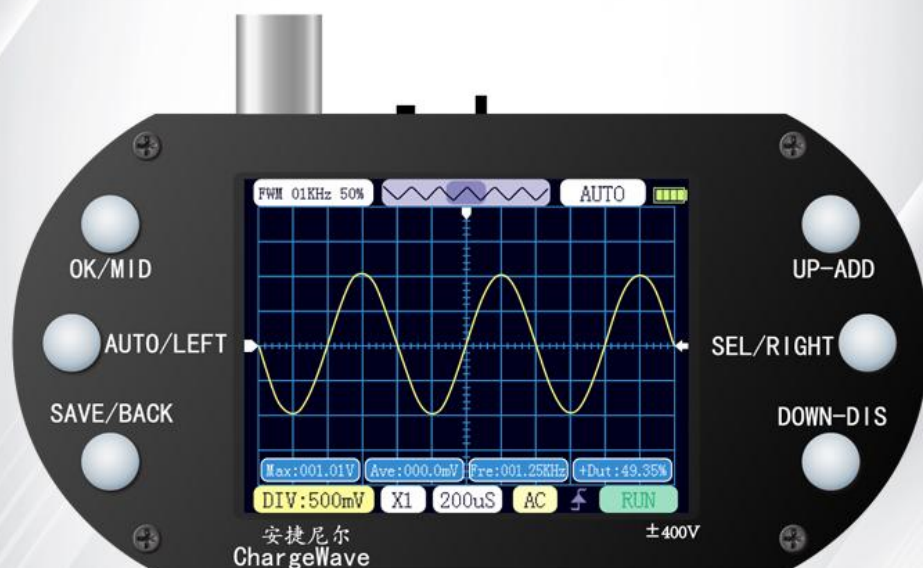


安捷尼尔
ENGINEER

SMALL MULTI-FUNCTION OSCILLOSCOPE

Mini | Functional | Composite Functions | Cost-effective



SUPER PERFORMANCE
CAN'T BE REFUSED THE ADVANTAGE

≡ SETTING PARAMETERS ≡

Model number	SCO_1_500K
Rate of adoption	2.5MHz
Analog bandwidth	500KHz
Actual bandwidth	200KHz
Measuring range	X1/±40V X10/±400V
Input impedance	1MΩ
Depth of storage	1Kb
Parameter type	12
PWM output	frequency 1K-80KHz Duty 0-100% Voltage 3.3V
Function	17 Waveforms
Output	Frequency 1~1KHz Voltage range 0~3.3V (fix)
Display	2.4" TFT 320*240Resolution ratio
Charging voltage	5V
Battery capacity	600mA/h 3.7V
Upper computer	Not supported for now
Firmware upgrade	Support
Size	Width 50cm and length 100cm(excluding BNC)

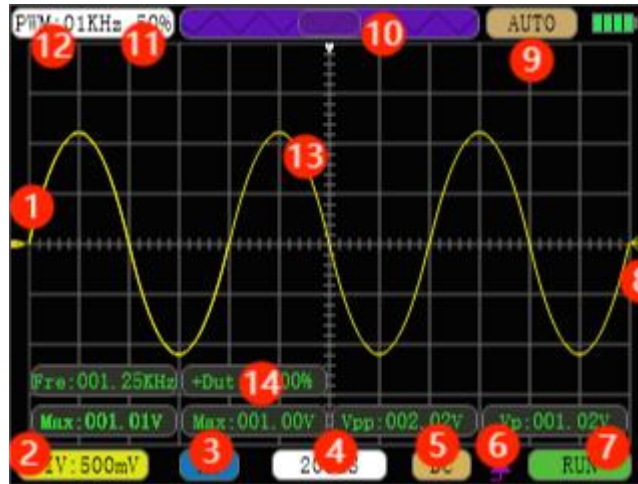
BUTTON FUNCTIONS AND PARAMETERS

≡ Button instruction ≡

Button	Short press	Touch & Hold	Save the key description
OK/MID	Confirm	Center the arrow	Open the corresponding saved waveform
AUTO/LEFT	Automatic adjustment	Automatic calibration interface	Press and hold to delete the image
SAVE/BACK	Waveform save/return	Go to the Save screen	Tap to exit the screen
UP-ADD	Up /Increase	Continuous +	+
SEL/RIGHT	Switching categories	Go to the menu page	Press and hold to delete the image
SDOWN-DIS	Decline/ Decrease	Continuous -	—

Tip: The key function is designed according to the habits of engineers. If the users have better suggestions, we can put forward their own suggestions. We will make corresponding upgrades and updates according to the needs of customers

Home Screen Introduction



Press SEL button to switch setting categories, and set parameters UP/DOWN

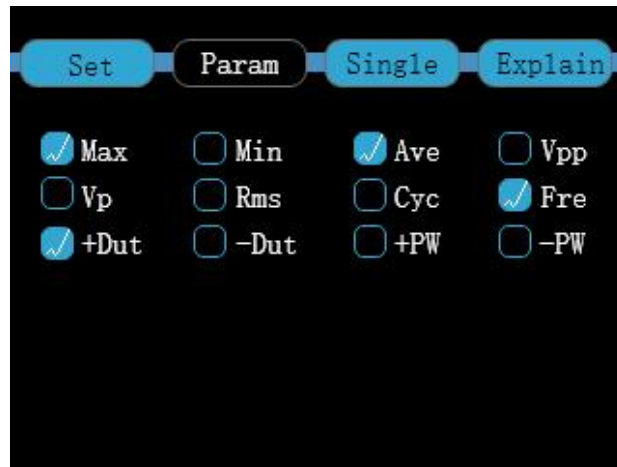
Main screen Description:

- 1.Where baseline 0 is located
- 2.Vertical component
- 3.X1/X10 Attenuation multiple(This position is only to switch the internal calculated value. The actual measurement can only be done by using the voltmeter pen with 10x attenuation. That is to say, if the voltmeter pen with 10x attenuation is not used for measurement, the maximum measurement can only be 40V)
- 4.Horizontal timebase
- 5.Coupling condition
- 6.Rising edge/falling edge
- 7.RUN/STOP
- 8.Vertical trigger position
- 9.Trigger mode(AUTO/Nomal/Single)
- 10.Horizontal trigger position
- 11.PWM output duty ratio (resolution 1%, independent of DAC output duty ratio)
- 12.PWM output frequency (resolution 1KHz, independent of DAC output frequency)
- 13.Corrugated display
- 14.Parameter display (Set the type of parameter displayed on the Settings page)

Other interface switching description:

- 1.On the home screen (above), hold down the SEL button to enter the menu screen. Hold down again to return to the menu screen
- 2.On the home screen (above), long press SAVE to enter the SAVE screen, short press Save to return

Introduction to the Settings menu page



Select the options that need to be set by pressing the button up and down, and press OK to modify

【1】 Scroll mode: The scroll mode is valid only when it is enabled and the mode is AUTO and the time base is greater than or equal to 100mS

【2】 Language: Currently, this product only supports switching between Chinese and English. If you need other languages, please feedback to us. We will continue to update in the future

【4】 Screen brightness, level 10 screen brightness, select, use the OK key to adjust

Section describes the Signal menu



1. Press the up or down button to switch the categories to be set. The selected categories will turn blue

2. Press the OK button to confirm the selection and the text will turn green

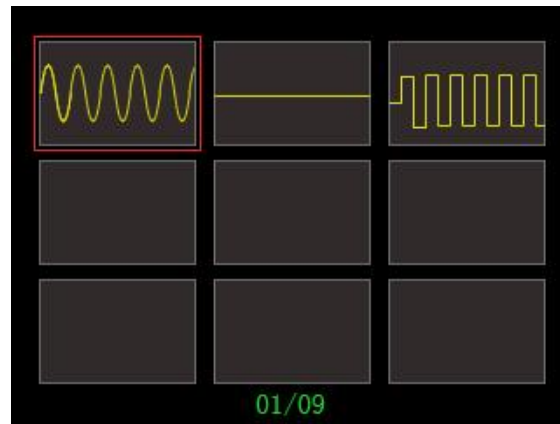
3. Press the up or down button to adjust the selected category

Tip:

a. The function generator of the oscilloscope can only output 17 different waveforms of 1Hz~1000Hz with a resolution of 1Hz

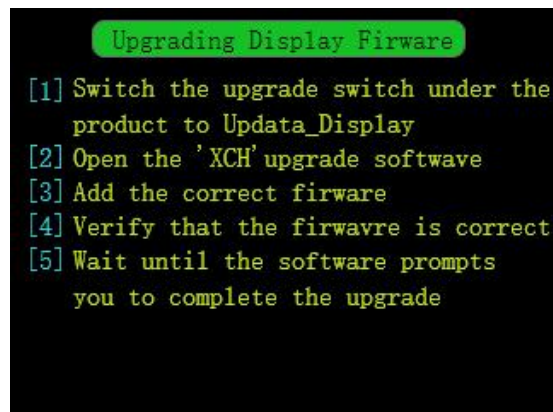
b. The output signal ranges from 0 to 3.3V and cannot be adjusted for the time being. The output signal can only be adjusted within this range even if the software is updated

Waveform saving interface



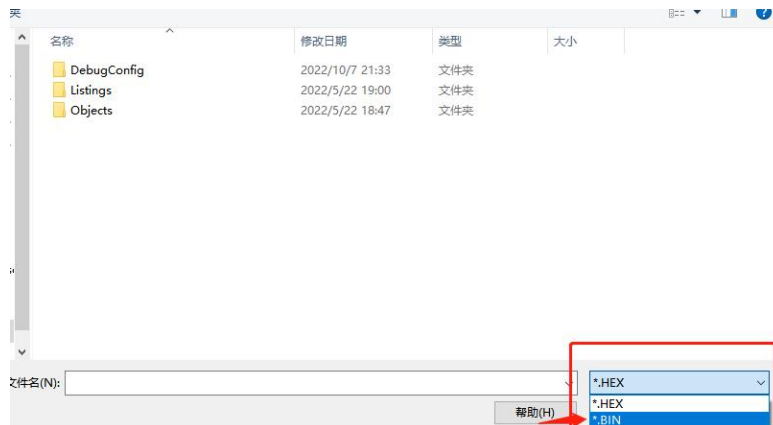
1. Select the saved waveform by pressing the up and down buttons, press the OK key to enlarge the display, and press again to return
2. Hold down the AUTO/SAVE button to delete the image
3. Customer interface optimization and other needs, please contact us for feedback, we will continue to update

Firmware Upgrade Description



Preparations for the display firmware upgrade

1. Press the OK button to start or reset the screen to enter the display firmware upgrade mode
2. Flip the toggle switch in the lower right corner to D(middle position)
3. Connect the Typec interface to the computer and open the corresponding software



Announcements

1. When measuring voltage over 40V, be sure to switch the probe gear to X10
2. When measuring high voltage (more than 36V AC voltage), do not touch the metal part of the instrument, especially the BNC interface position
3. Measure the mains. Try not to measure while charging, to avoid short circuit of the ground wire
4. The charging voltage should not exceed 6.8V to avoid burning the charging chip
5. When the firmware is not being upgraded, flip the switch in the lower left corner to Nomal

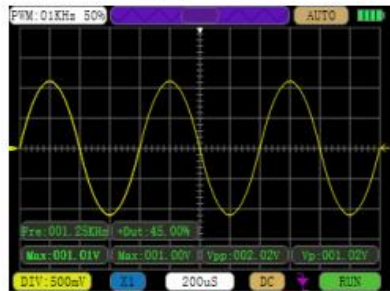
Other description

1. 2uS, 5uS cannot be used to measure the waveform of sine wave
2. Save the waveform. Only the displayed part of the current screen can be saved
3. Part of the measured signals cannot be correctly captured by AUTO, which requires manual adjustment, such as waveforms with frequencies lower than 10Hz

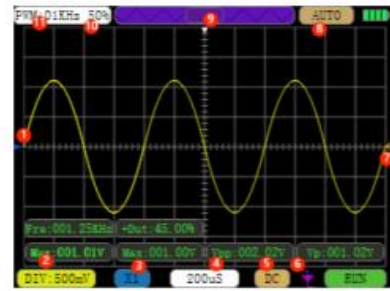
Important to remind again: the probe of free product distribution is X1 times straight through the probe, the maximum voltage can only be measured $\pm 40V$, if the customer needs to measure more than 40V, such as mains power $\pm 220V$ (Effective value, the actual maximum voltage is 311.8V), Additional voltmeter pens with X10 attenuation will need to be purchased

//Supplementary statement ↓

On this screen, press the SEL button to switch control categories. The selected categories will turn blue. After being selected, you can use the UP/DOWN button to perform operations



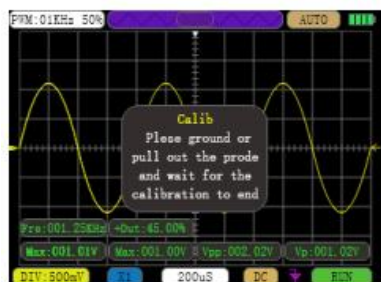
Main interface display



The baseline icon is selected

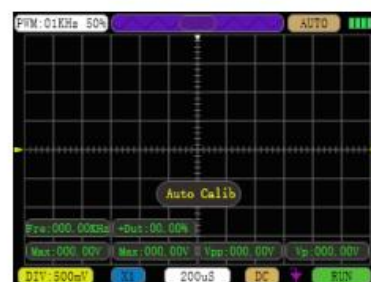
Description of each category:

1. Baseline arrow:	The position of baseline 0, that is, the position where the 0V/ probe is grounded and suspended. If not centered, baseline calibration is required
2. Vertical component:	Each cell represents the voltage value, as shown in the figure where the baseline is in the middle, and each cell above the baseline represents the voltage value of 500mV
3. X1/X10	This parameter is only used to adjust the digital multiple of the parameter, which needs to be used with the X10 switch of the high voltage probe
4. Horizontal time base:	The time in the horizontal direction of each cell, as shown in the figure, is 200uS
5. Coupling mode:	AC coupling: The internal circuit isolates the DC component of the measured signal DC coupling: The DC component is not isolated
6. Rising/falling edge	For example, the triggering mode of falling edge is shown in the figure, and the triggering position is the intersection of 7 and 9. At this time, the signal is falling, and the triggering position can be modified by changing 7 and 9
7. Vertical trigger arrow	Change the vertical position of the trigger by adjusting the arrow position
8. Trigger mode	AUTO: Automatically triggered. Even if the signal is not triggered, it will be automatically displayed NORMAL: Normal trigger. In this mode, if the signal is not triggered, the display waveform will not change, and it will refresh again when triggered again SINGLE: Single trigger. In this mode, the oscilloscope waits and triggers the signal only once
9. Horizontal trigger arrow	Change the horizontal position of the trigger by adjusting the arrow position (built-in 1K storage depth)
10. Duty cycle	Duty ratio of output PWM signal (independent of signal duty ratio of output DDS)
11. Frequency	The frequency of the output PWM signal (independent of the signal frequency of the output DDS)



Calibration interface display

Press the AUTO button to access the screen

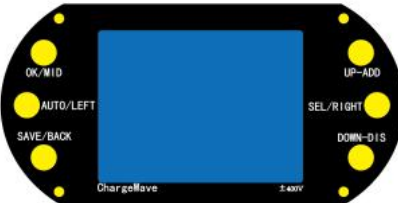


Calibration in progress

Press the OK button to start

Tip: During calibration, the probe had better be pulled out or grounded. Do not touch the probe or BNC port during calibration to avoid interference resulting in calibration failure. If the calibration fails, it can be calibrated again until the yellow signal line is almost equal to the arrow (there may be an error of one or two display points, which is a normal phenomenon).

Each key on the panel is illustrated

OK/MID 1. On the oscilloscope display screen, tap to pause or start 2. When the arrow target is selected (the arrow on the screen turns blue is selected), press it to center the arrow 3. On the menu screen, press the Confirm button	UP-ADD DOWN-DIS 1. In the oscilloscope display interface, according to the different categories selected, can be added or subtracted, up, down, left and right 2. Move the option box when the menu is displayed	
AUTO/LEFT 1. In the oscilloscope display interface, press it to enter automatic adjustment. According to the state of the measured waveform, the oscilloscope measurement conditions will be automatically adapted, and the waveform will be displayed in a clear appearance on the interface 2. Press the button, the automatic calibration window will pop up, and press the OK button to enter the automatic calibration 3. On the menu screen, switch the menu to the left 4. On the save screen, press to delete the saved image		SEL/RIGHT 1. In the oscilloscope display interface, press down to switch the setting categories. The selected categories will turn blue 2. Press the key to enter or exit the menu 3. On the menu screen, switch the menu to the right 4. On the save screen, press to delete the saved image
SAVE/BACK 1. On the oscilloscope screen, press to save the screenshot 2. Press and press to enter the Save screen, where you can view the waveform saved by yourself. Press the OK button to enlarge the view 3. On the SAVE screen, press to exit the screen and return to the oscilloscope display screen		

220V AC measurement tutorial and matters needing attention

Note: 1. It is absolutely prohibited to measure the mains electricity ! ! ! ! ! while charging

Note: 2. The measurement process must not touch any metal parts of the oscilloscope!!!

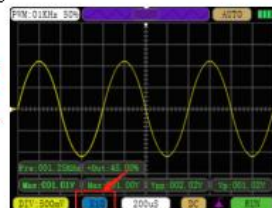
Step 1: High voltage probe toggle to X10 end



Step 2: Oscilloscope open in X10 multiples



On this screen, press the SEL key to switch to X1, and then press UP or Down



Step 3: The negative end of the high voltage probe is connected to the zero line, and the positive end is connected to the live line

Positive interface



Negative end interface

